

A Pilot Study of PE-RankFormer: Context-Conditioned Relational Transformer Learning for Prime-Editing Efficiency Prediction

Research Proposal

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Important scope note. The word “pilot” refers to *model scope*, not data scope. The study will assemble and use the **complete 297,962-example training corpus** used by OptiPrime, including the 74,769 new OptiPrime measurements and the compatible measurements drawn from the three prior studies. The pilot deliberately avoids a broad architecture search. Instead, it tests one model framework selected to have a high probability of success, with only a small number of ablations and hyperparameter choices.

Executive Summary

OptiPrime currently reports the strongest published accuracy for prime-editing (PE) efficiency prediction. Its main methodological innovation is a strong mechanistic inductive bias: separate learned pseudorates are assigned to predefined biochemical steps and integrated through a reaction model. This proposal tests the complementary hypothesis that, given the same large heterogeneous training corpus, a sufficiently structured but nonmechanistic end-to-end learner can equal or exceed OptiPrime’s predictive accuracy.

We propose **PE-RankFormer**, a compact context-conditioned relational Transformer. The model will learn directly from: (i) an aligned representation of the unedited and edited target sequence, (ii) the spacer/PBS/RTT pegRNA sequence with segment identity, and (iii) experimental context tokens such as cell type and prime-editor configuration. It will use cross-attention to learn target–pegRNA interactions. The key training innovation is a joint objective that combines absolute efficiency regression with **within-target ranking**, directly optimizing the practical decision problem: among several candidate pegRNAs for the same edit, which ones should be tested first?

The complete OptiPrime corpus contains 297,962 PE measurements across 40 experimental contexts. OptiPrime used protospacer-stratified fivefold cross-validation and reported mean Pearson $r = 0.693$ and Spearman $\rho = 0.745$ on its cross-validation benchmark, compared with substantially lower correlations for DeepPrime-FT and PRIDICT2.0 on the newly collected OptiPrime screens [1]. The proposed pilot will use the same leakage-control principle, evaluate both global and within-target ranking metrics, and use the official implementations of OptiPrime, DeepPrime-FT, and PRIDICT2.0 where possible.

A single NVIDIA RTX PRO 6000 Blackwell workstation GPU provides 96 GB of GDDR7 ECC memory [7]. Because the relevant sequences are short and the proposed network is only about 20–30 million parameters, the project is comfortably feasible on one card. We estimate roughly 1.5–3.5 GPU-hours per fold after data caching and compilation, with the complete fivefold evaluation expected to require roughly 8–18 GPU-hours for the primary model; an initial feasibility run should be available after a single fold.

1 Scientific Hypothesis and Pilot Objective

The primary hypothesis is:

A modern end-to-end model can outperform a strongly mechanism-constrained model if it is trained on the same complete heterogeneous dataset, explicitly represents the relation among unedited target, edited target, and pegRNA, conditions on experimental context, and optimizes the within-target ranking task in addition to global regression.

The pilot is designed to answer three questions:

1. **Representation:** Does a relational sequence representation recover the information that OptiPrime obtains through hand-selected mechanistic modules?
2. **Context:** Can one jointly trained model learn across the 40 experimental contexts without context-specific fine-tuning?
3. **Objective:** Does ranking-aware training improve selection of the best pegRNAs, especially when candidate designs are all reasonably efficient?

The pilot is intentionally **not** a large model-selection exercise. If the proposed architecture fails to approach OptiPrime on a full-data fold, the project can be stopped or redesigned before extensive GPU use.

2 Proposed Model Framework

2.1 Design principle

The model should have a weak but useful structural inductive bias: it should know which nucleotides belong to the target, the desired edit, the spacer, the PBS, and the RTT, but it should **not** be told a biochemical reaction graph or which sequence properties control which mechanistic step. Thus the model learns the mapping

$$\hat{E} = f_{\theta}(S_{\text{WT}}, S_{\text{edit}}, S_{\text{spacer}}, S_{\text{PBS}}, S_{\text{RTT}}, C), \quad (1)$$

where C contains experimentally known context variables.

2.2 Input representation

Each experiment is converted into three coordinated inputs.

1. Aligned edit sequence. The unedited and desired edited sequences are globally aligned in a fixed local target window. Each position is encoded as a *paired-base token*

$$(b_{\text{WT}}, b_{\text{edit}}), \quad b \in \{A, C, G, T, -\}.$$

This produces at most 25 elementary edit states and makes substitutions, insertions, and deletions explicit without providing a mechanistic interpretation. Position embeddings and region flags indicate protospacer, PAM, nick position, edit interval, and homology-arm boundaries when known.

2. pegRNA sequence. The spacer, PBS, and RTT are encoded at single-nucleotide resolution. Segment embeddings mark SPACER, PBS, and RTT. The model can therefore infer lengths, GC composition, edit distance, complementarity, and sequence motifs directly from sequence rather than from hand-engineered thermodynamic features.

3. Experimental context. Categorical tokens encode variables that are known before the experiment and are available in the pooled training data: cell line, PE system/editor, MMR-modulation condition (for example PE2 versus PE4 when applicable), pegRNA scaffold/epgRNA status, nuclease/PAM variant, and study/assay identity when required to harmonize batch-specific scales. Missing fields use explicit UNK categories rather than imputation from outcomes.

2.3 Architecture: PE-RankFormer

The proposed pilot model has four components (Fig. 1).

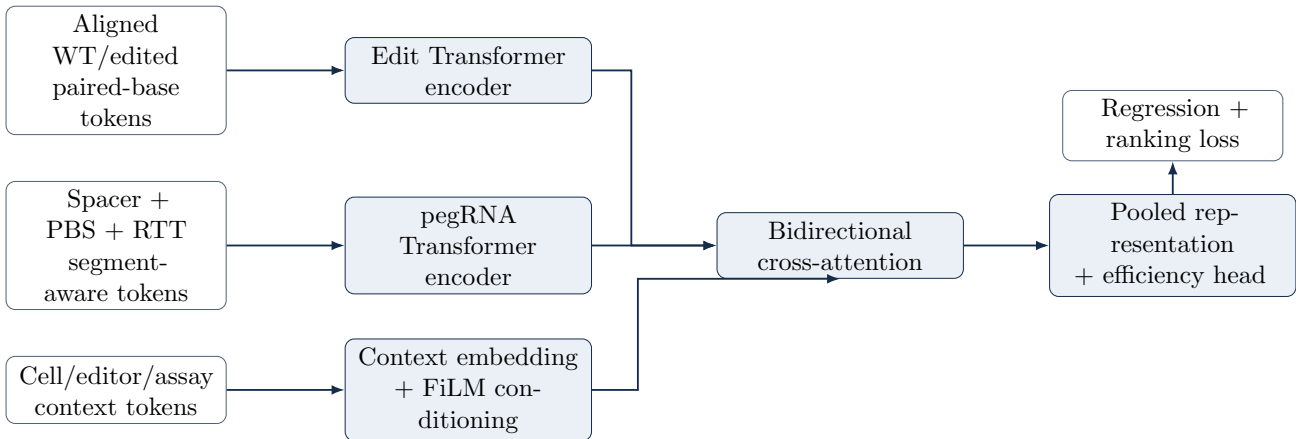


Figure 1: Pilot PE-RankFormer architecture. The structural bias specifies only the identities and alignments of the experimental objects. No predefined PE reaction graph or pseudorates are imposed.

Edit encoder. A 5–6 layer Transformer ($d = 384$, 6 attention heads, feed-forward width 1,536) encodes the paired WT/edited tokens.

pegRNA encoder. A 4-layer Transformer of the same hidden width encodes spacer/PBS/RTT tokens with segment embeddings.

Relational cross-attention. Two bidirectional cross-attention blocks allow edit positions and target bases to attend directly to pegRNA positions and vice versa. This is intended to learn PBS-target complementarity, RTT-edit correspondence, spacer-target compatibility, and other interactions without specifying their biochemical consequences.

Context conditioning. Context embeddings modulate hidden states using feature-wise linear modulation (FiLM),

$$\text{FiLM}(h; c) = \gamma(c) \odot h + \beta(c), \quad (2)$$

which is substantially simpler and more stable than a mixture-of-experts model for a first pilot while still allowing the same sequence to behave differently under different PE systems or cell contexts.

Prediction head. Attention pooling followed by a two-layer MLP outputs a scalar efficiency. A sigmoid constrains the final prediction to $[0, 1]$. The expected model size is approximately 20–30 million parameters, small enough to train comfortably from scratch on 297,962 labeled examples.

2.4 Training objective: regression plus within-target ranking

A major weakness of global correlation is that a model can perform well by learning that one target is generally easy and another target is generally difficult, while failing to rank pegRNAs within either target. To align training with pegRNA selection, we use

$$\mathcal{L} = \mathcal{L}_{\text{reg}} + \lambda \mathcal{L}_{\text{rank}}. \quad (3)$$

For regression, efficiency is stabilized with a clipped logit transform and optimized with Huber loss. For ranking, pairs are sampled only among pegRNAs that install the same edit under the same experimental context. If $E_i > E_j$, a RankNet-style term is

$$\ell_{ij} = \log(1 + \exp[-(s_i - s_j)]), \quad (4)$$

where s_i and s_j are pre-sigmoid model scores. To avoid domination by near-ties within experimental noise, pairs with $|E_i - E_j|$ below a small threshold (initially 2 percentage points) receive zero or reduced weight. The pilot will compare $\lambda = 0$ with one prespecified ranking weight (for example $\lambda = 0.25$); this is the main ablation, not a broad search.

3 Training Data: Full 297,962-Experiment Corpus

3.1 Data sources

OptiPrime was trained jointly on 297,962 PE experiments spanning 40 experimental contexts [1]. The corpus combines its new paired pegRNA–target screens with three prior studies: PRIDICT [3], DeepPrime [2], and PRIDICT2.0/ePRIDICT [4].

Source	Role	Acquisition plan
OptiPrime	74,769 new measurements	Use processed efficiencies and full design sequences from Supplementary Tables 4–5; the uploaded workbook already contains the nonmissing editing-efficiency labels. Raw SRA reads are reserved for audit, not routine model construction.
PRIDICT (2023)	Prior-study data	Download processed supplementary/GitHub tables and reproduce OptiPrime-compatible filtering.
DeepPrime (2023)	Prior-study data	Assemble processed pegRNA-target measurements from the public study data and DeepPrime release; harmonize editor/cell metadata.
PRIDICT2.0 (2025)	Prior-study data	Use processed supplementary/repository data; retain only experiments included by the OptiPrime training-data definition.

Table 1: Full-data assembly plan. The exact retained rows from the three prior studies must sum to 223,193, because $297,962 - 74,769 = 223,193$.

3.2 Reconstructing the exact OptiPrime corpus

The primary data-engineering deliverable is a versioned table with one row per PE experiment and the following canonical schema:

$$\{\text{study, context, target_id, protospacer, } S_{WT}, S_{edit}, S_{spacer}, S_{PBS}, S_{RTT}, \text{editor, cell, } E\}.$$

The exact row selection should be recovered from the OptiPrime public data loader/training code and the source studies, rather than guessed. Quality checks will include:

- exact total of 297,962 retained measurements;
- exact total of 74,769 OptiPrime-new measurements;
- no train/test sharing of protospacer groups;
- canonical DNA orientation and consistent edit alignment;
- efficiency units standardized to fractions in $[0, 1]$;
- duplicate audit across studies using protospacer, target sequence, edit, context, and pegRNA sequence hashes;
- explicit missing-data masks for context fields rather than outcome-based imputation.

3.3 Why raw SRA reprocessing is not required for the pilot

The goal is predictive modeling, not re-estimation of sequencing counts. Where processed per-design efficiencies are publicly available, those values should be treated as the training labels used by the original studies. Raw SRA data are useful only for resolving missing or ambiguous records. This reduces unnecessary preprocessing and minimizes the chance that small differences in CRISPResso or read-filter settings create labels that differ from the published models.

4 Evaluation Design

4.1 Leakage-resistant cross-validation

The primary evaluation will mirror OptiPrime’s key leakage-control rule: **fivefold cross-validation stratified/grouped by protospacer sequence** [1]. All measurements sharing a protospacer are assigned to the same fold, including measurements from different contexts. If the exact published fold assignments are available in the released code, they will be reused. Otherwise, a deterministic grouped split with a fixed public seed will be generated, and all runnable baselines will be evaluated on the same held-out rows.

For the initial feasibility gate, only one locked fold is trained. If the model reaches the prespecified threshold below, the remaining four folds are run without changing the architecture.

4.2 Primary comparison set

The most direct benchmark is the 74,769 measurements generated in the OptiPrime study because published cross-validation comparisons are available on these data. OptiPrime reports mean $r = 0.693$ and $\rho = 0.745$ under protospacer-stratified fivefold cross-validation; on the same new datasets, the reported retrospective values are approximately $\rho = 0.399$ for DeepPrime-FT, 0.538 for PRIDICT2.0-HEK, and 0.590 for PRIDICT2.0-K562 [1].

The primary pilot comparison is therefore:

PE-RankFormer vs. OptiPrime vs. DeepPrime-FT vs. PRIDICT2.0-HEK.

PRIDICT2.0-K562 will also be included when directly applicable. OPED and DTMP-Prime are useful secondary Transformer baselines, but they are not required for the initial go/no-go decision because they were trained under different data regimes [5, 6].

4.3 Metrics

Primary metric. Mean Spearman rank correlation ρ on held-out examples, reported overall and separately for each of the four new OptiPrime conditions.

Secondary global metrics. Pearson r , MAE, RMSE, and calibration slope/intercept.

Design-centric metrics. For each target/edit/context group with enough candidate pegRNAs:

- within-target Spearman ρ_{within} ;
- NDCG@5;
- top-1 regret: $E_{\text{max}} - E_{\text{model choice}}$;
- top-3 capture: whether at least one of the experimentally top-3 pegRNAs appears in the model top-3 or top-5;

- achieved best efficiency if the experimenter tests the model’s top k designs, $k \in \{1, 3, 5\}$.

These metrics directly test whether the model solves the experimental selection problem rather than merely learning between-target differences.

External stress test. If the published arrayed ATP1A3 data can be reconstructed cleanly, they will remain completely untouched during model development and serve as a small external test. OptiPrime reported site-specific rank correlations ranging from 0.196 to 0.604 on four ATP1A3 mutations [1], leaving room for a purely learned model to demonstrate improved within-locus ranking.

4.4 Statistical comparison

Uncertainty will be estimated by bootstrap resampling at the protospacer/target group level rather than at the individual-row level. Differences in correlation on the same observations can additionally be tested with Steiger-type dependent-correlation tests; design-centric metrics will use grouped bootstrap confidence intervals. The pilot should emphasize effect size and stability across folds rather than extremely small P values produced by hundreds of thousands of observations.

5 Pilot Execution Plan

5.1 Stage 0: data audit and frozen split

1. Reconstruct the exact 297,962-row corpus.
2. Generate canonical sequence alignments and context dictionaries.
3. Reproduce counts by study and context.
4. Freeze one protospacer-disjoint pilot fold and the remaining four CV folds.
5. Cache tokenized records to avoid repeated preprocessing.

5.2 Stage 1: one-fold feasibility run

Train the proposed PE-RankFormer from scratch on four partitions and evaluate on one. Use BF16 mixed precision, fused scaled-dot-product attention/FlashAttention where available, AdamW, gradient clipping, cosine or one-cycle learning-rate decay, and early stopping on validation Spearman correlation. A reasonable initial schedule is 20–30 epochs with effective batch size 512–1,024.

Only two models are required in Stage 1:

1. PE-RankFormer with regression loss only ($\lambda = 0$);
2. PE-RankFormer with regression + within-target ranking loss.

This isolates whether the ranking formulation adds value while avoiding an expensive architecture search.

Stage-1 feasibility criterion. Proceed to full fivefold evaluation if the ranking-aware model (i) trains stably on the full corpus, (ii) achieves held-out Spearman $\rho \geq 0.72$ on the OptiPrime-new subset or is within 0.02 of the corresponding OptiPrime fold, and (iii) improves at least one design-centric metric over the regression-only model by a practically meaningful margin (target: $\geq 5\%$ relative improvement in top-3 capture or top- k regret).

5.3 Stage 2: complete fivefold pilot

Without changing the architecture, train the remaining four folds. Report mean $\pm$ standard deviation across folds and compare to the published/official baseline predictions. Only after this stage should larger models, pretrained DNA encoders, mixture-of-experts routing, or setwise candidate modeling be explored.

6 Single-GPU Feasibility and Compute Budget

The NVIDIA RTX PRO 6000 Blackwell Workstation Edition has 96 GB GDDR7 ECC memory [7]. This is substantially more memory than required for the proposed 20–30M parameter model with sub-256-token inputs.

Component	Expected re-source	Planning estimate
Model parameters	20–30M	Well below GPU-memory limits; train from scratch.
Input length	typically < 256 tokens after structured encoding	No need for long-context architectures.
Precision	BF16	Native Blackwell mixed-precision path; retain FP32 optimizer statistics as needed.
Effective batch	512–1,024	Use direct batching or modest gradient accumulation.
Steps per epoch	roughly 300–600	Depends on final batch size and fold size.
Epochs	20–30 maximum	Early stopping expected before maximum.
One CV fold	about 1.5–3.5 GPU-hours	Conservative engineering estimate; benchmark first 1,000 steps and update.
Full 5-fold primary model	about 8–18 GPU-hours	Can be completed sequentially on one card.
Regression-only ablation	about 1.5–3.5 GPU-hours for pilot fold	Only one ablation required initially.
Total pilot GPU budget	about 12–25 GPU-hours	Includes reruns/checkpoints and baseline inference margin.

Table 2: Compute plan for a single RTX PRO 6000 Blackwell. Times are engineering estimates rather than guarantees; the first 1,000 training steps will be timed to replace them with measured throughput.

The 96 GB memory capacity makes it unnecessary to use parameter offloading, model parallelism, or aggressive checkpointing. The dominant practical issues are more likely to be data harmonization and grouped ranking batches than GPU capacity.

7 Minimal Hyperparameter Policy

To keep the pilot scientifically clean and computationally modest, hyperparameters will be mostly fixed before examining the final held-out fold:

- hidden dimension: 384;
- edit encoder: 6 layers; pegRNA encoder: 4 layers; cross-attention: 2 blocks;
- attention heads: 6;
- FFN width: 1,536;
- dropout: 0.10;
- AdamW, weight decay 0.01;

- peak learning rate selected from only $\{1 \times 10^{-4}, 3 \times 10^{-4}\}$ on an internal validation split;
- ranking weight selected from $\{0, 0.25\}$ for the pilot;
- early stopping based on validation Spearman ρ with a minimum number of warm-up epochs.

The objective is to test the framework, not to obtain a benchmark gain through extensive hyperparameter tuning.

8 Risks and Mitigations

Risk	Mitigation
Exact 297,962-row corpus is difficult to reconstruct	Treat data reconstruction as Stage 0; use OptiPrime loaders/source-study supplements; require exact row-count reconciliation before model training.
Strong cross-study scale/batch effects	Use explicit context/study tokens; report per-context metrics; keep a no-study-token sensitivity analysis for the final study if necessary.
Transformer learns target difficulty but not pegRNA ranking	Use within-target ranking loss and design-centric metrics as primary methodological safeguards.
Ranking loss overfits noisy near-ties	Downweight/omit pairs separated by less than 2 percentage points; test sensitivity only if needed.
Model underperforms OptiPrime	Pilot still answers the scientific question cheaply; inspect whether gap is global calibration or within-target ranking before adding complexity.
Pretrained DNA language model might help	Defer to Stage 3. With nearly 300k supervised labels and short relational inputs, training from scratch is the cleanest first test.

9 Go/No-Go Decision and Follow-up

The project should be considered technically feasible if the full-data model trains stably within the expected single-GPU budget and reaches performance close to OptiPrime on the first locked fold. It should be considered scientifically promising if the fivefold mean either:

1. exceeds OptiPrime’s reported new-screen rank correlation ($\rho \approx 0.745$) by a meaningful margin (target ≥ 0.02 absolute), or
2. matches global correlation while clearly improving within-target ranking/top- k selection metrics.

The second outcome is potentially more important scientifically: a model that selects better pegRNAs while having similar global ρ solves the experimental design problem more directly.

If the pilot succeeds, the next stage can test exactly one extension at a time: (i) setwise attention among candidate pegRNAs for the same target, (ii) context-conditioned mixture-of-experts, (iii) pretrained genomic encoders such as Nucleotide Transformer, or (iv) endogenous chromatin features for locus-specific prediction. These should be treated as follow-up studies, not requirements for demonstrating the core feasibility of PE-RankFormer.

Expected Outcome

This pilot is designed to provide a decisive answer with modest compute. It uses the **full public training corpus**, a model that is expressive but small enough for rapid iteration, an input representation aligned with the mathematical structure of the editing problem rather than a predefined biochemical mechanism, and an evaluation protocol that measures both global prediction and the practical task of choosing the

best pegRNA. A positive result would justify a larger study and support the broader conclusion that strong mechanistic constraints are not required for state-of-the-art PE efficiency prediction when data heterogeneity and within-target ranking are modeled directly.

References

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